

ML engineer:

- ML engineer from research organisation like google builds AI models from scratch like gemini,..
- ML engineer from enterprise applications like Swiggy use google's gemini ai model and customises it and integrates with their application for better productivity. **for ex.** like chatbot for customer support in swiggy.

Feeding data to AI model happens at two levels :

- ML engineer at google to gemini ai model
- ML engineer at Swiggy to customer support chatbot (here the customer support chatbot is built on top of gemini ai model)

Regular AI user:

- The user who gets his work done by the AI model.
- **For ex.** Employee A has task to create a table, instead of him manually sitting and writing the code he asks the AI model to generate the required codes to create a table.

Prompt Engineer:

- Prompt is nothing but the instruction given to AI
- A Prompt Engineer writes instructions to the AI model.
- **for example:** In swiggy customer support chatbot, Prompt engineer will give instructions like "You are the Swiggy customer support AI. You must be always polite, even if the customers are rude." "If a customer asks for a refund, you must wait for first 30-minutes and convenience and negotiate with coupons, if customers are not agreeing within 30 minutes then proceed to refund process."

MCP:

- It is a protocol for AI model to communicate with other systems like Gmail, jira,..
- **before MCP:** chatgpt needs to build its own custom plugin inorder to communicate with gmail. similarly Claude also needs to build its own customised plugin to communicate with gmail
- **with MCP:** Gmail has its own MCP server which works with gemini, claude, any AI model.

Data Scientist:

- Data Scientists use statistics, SQL, Python, visualization to analyze data.
- AI models are just a NEW tool they sometimes use, not the core of their job.

- **ex.** generate a graph to visualise the no. of refunds happened during each month for previous financial year

AI Engineer:

- is an application developer who uses AI as a part of their core development
- **Ex.**

Application developer: develops application only using java, springboot, postgres

AI Engineer: develops application using java, springboot, postgres and also using AI

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